DOC327 Working With Data

LECTURE 02

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Why do Organizations Need Data?

[Continued from previous lecture]

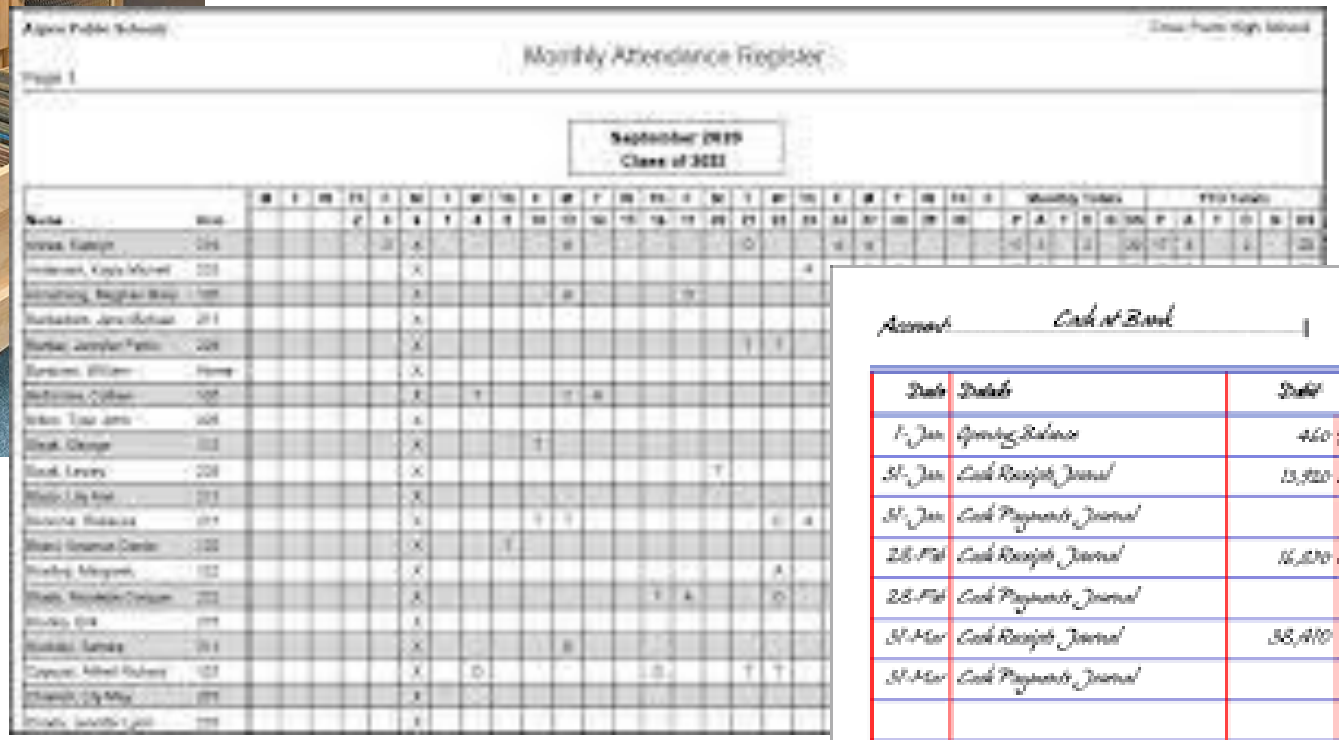
- **Increased Productivity/Improved Performance**
- **Better & Quick decision-making**
- **Customer relationship management**
- **Be Strategic In Your Approaches**
- **Get The Results You Want**

How did organizations start keeping their data?

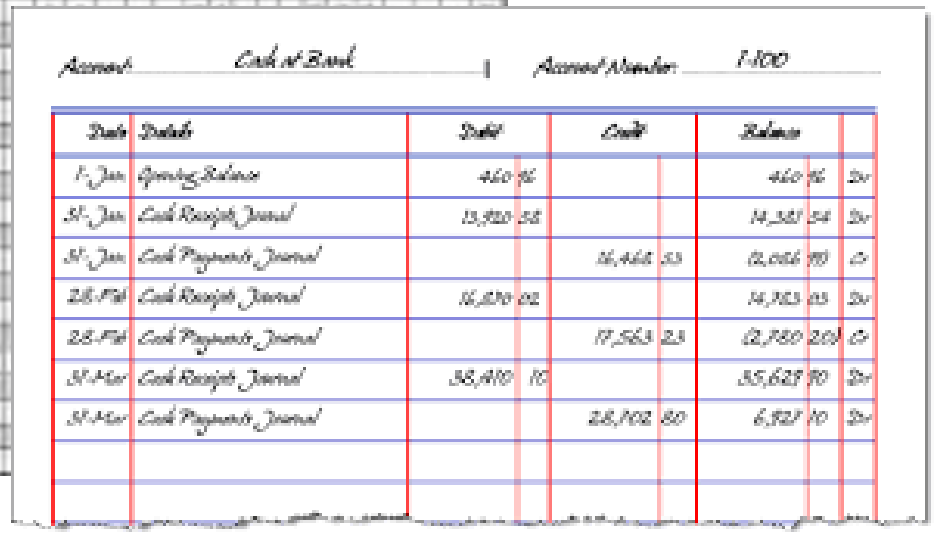
- Organizations started to collect data and keep a permanent record of data with them. (Data type depended on the operations they were performing in their business.)
- They started arranging these data using files.
- Earlier data storage methods in organizations.
  - File Cabinets
  - Registers
  - Ledgers

How did organizations start keeping their data? [Continued..]





Monthly Attendance Register



Account: Cash at Bank | Account Number: 1-100

Date	Details	Debit	Credit	Balance
1st Jan	Opening Balance	420 56		420 56 Dr
5th Jan	Cash Receipts Journal	13,930 58		14,351 54 Dr
5th Jan	Cash Payments Journal		16,466 53	(2,066 99) Cr
25th Feb	Cash Receipts Journal	16,870 02		14,803 03 Dr
28th Feb	Cash Payments Journal		17,563 23	(2,760 20) Cr
31st Mar	Cash Receipts Journal	38,410 10		35,628 90 Dr
31st Mar	Cash Payments Journal		28,702 80	6,926 10 Dr

Difficulties Faced With Manual File Based Systems

- Lack of security
- Prone to damage
- Limited access and collaboration
- High Costs
- Need higher storage space



Electronic File Based Systems

- Everyone needs a way to collect and have a permanent record of information that is important to them.
- Organizations used files for this purpose.
- With the introduction of computers, the concept of **"electronic collections of files"** came into being.
- **File-based systems were born as an attempt to computerize the traditional manual filing system.**

Electronic File-Based Systems [Continued..]

- What is a file-based system?
  - A file system is a technique of arranging files in a storage medium like a hard disk, pen drive, DVD, etc. It helps you to organize the data and allows easy retrieval of files when they are required. It mostly consists of different types of files like mp3, mp4, txt, doc, etc. that are grouped into directories. A file system enables you to handle the way of reading and writing data to the storage medium.
  - A system consisting of a collection of application programs; each of them is designed to manipulate data files.

Electronic File-Based Systems [Continued..]

- Each program defines and manages its own data
- Developed for more efficient data access but has disadvantages.

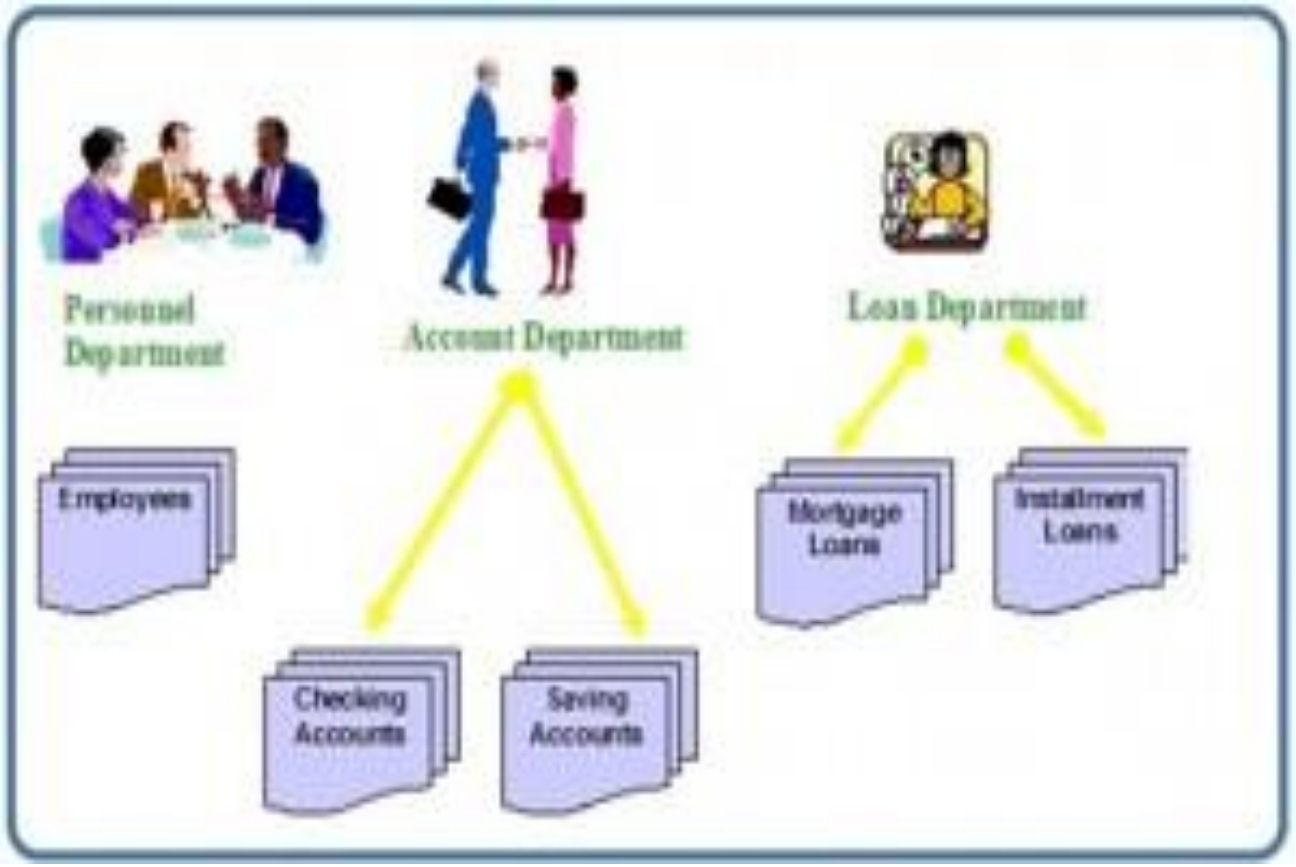


Electronic File-Based Systems [Continued..]

Consider a traditional banking system that uses the file-based system to manage the organization's data shown in Figure.

As we can see, there are different departments in the bank. Each has its own applications that manage and manipulate different data files.

For banking systems, the programs may be used to debit or credit an account, find the balance of an account, add a new mortgage loan, and generate monthly statements.



Disadvantages of Electronic File-Based Systems

- **Data Redundancy**
  - Often, within an organization, files, and applications are created by different programmers from various departments over long periods of time. This can lead to data redundancy, a situation that occurs in a database when a field needs to be updated in more than one table. This practice can lead to several problems such as:
    - Inconsistency in data format
    - The same information is kept in several different places (files)
    - Data inconsistency, a situation where various copies of the same data are conflicting, wastes storage space, and duplicates effort.

Disadvantages of Electronic File-Based Systems [Continued]

- **Integrity problems**
  - Problems with data integrity are another disadvantage of using a file-based system. It refers to the maintenance and assurance that the data in data storage are correct and consistent. Factors to consider when addressing this issue are:
    - Data values must satisfy certain consistency constraints that are specified in the application programs.
    - It is difficult to make changes to the application programs in order to enforce new constraints.

Disadvantages of Electronic File-Based Systems [Continued]

- **Security problems**
  - Security can be a problem with a file-based approach because:
  - There are constraints regarding accessing privileges.
  - Application requirements are added to the system in an ad-hoc manned security problems
  - Security can be a problem with a file-based approach because:
    - There are constraints regarding accessing privileges.
    - Application requirements are added to the system in an ad-hoc manner, so it is difficult to enforce constraints', so it is difficult to enforce constraints.

Disadvantages of Electronic File-Based Systems [Continued]

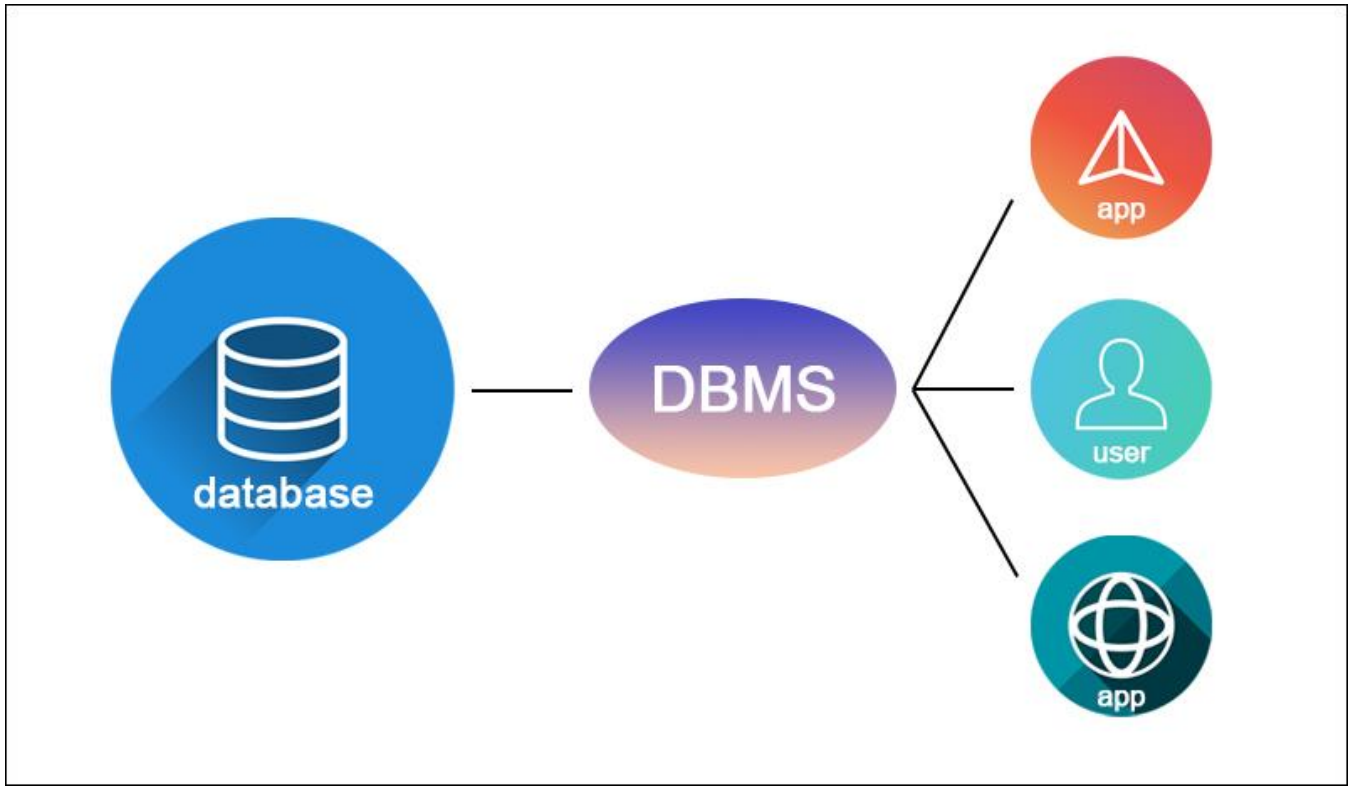
- **Concurrency access**
  - Concurrency is the ability of the database to allow multiple users access to the same record without adversely affecting transaction processing. A file-based system must manage, or prevent, concurrency by the application programs. Typically, in a file-based system, when an application opens a file, that file is locked. This means that no one else has access to the file at the same time.

Disadvantages of Electronic File-Based Systems [Continued]

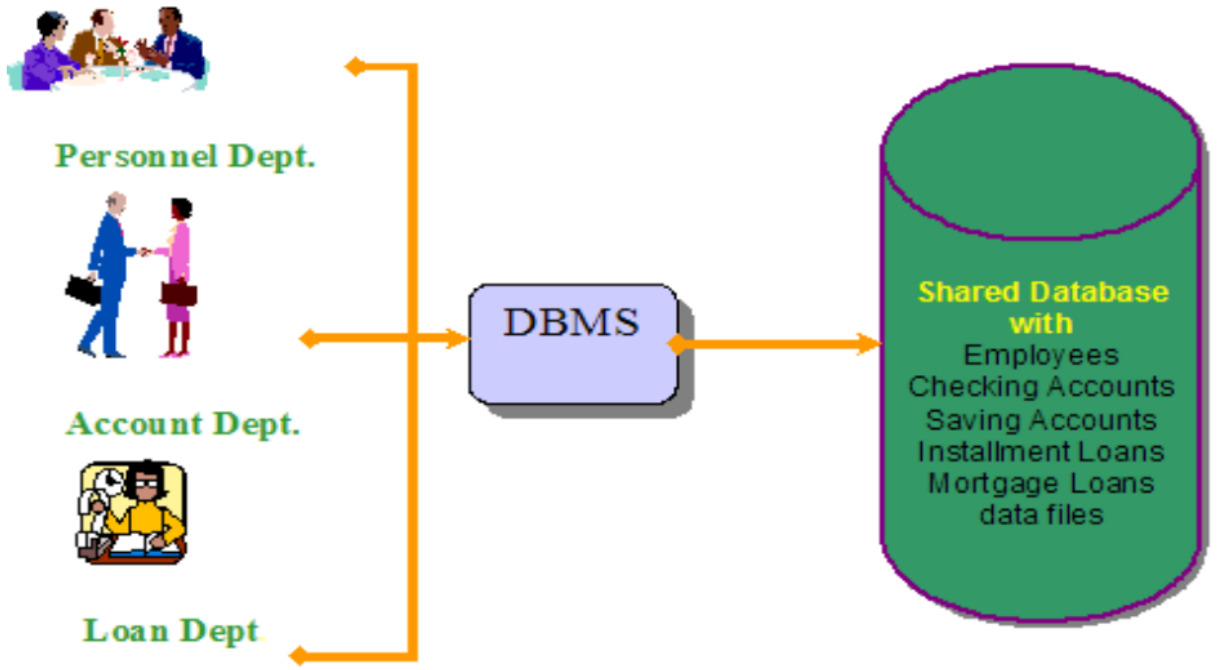
- Data definition is embedded in the application program.
- No controls of data beyond what the program specifies.
- **The solution was the database / DBMS approach.**

Database Management Systems

- What are Database Management Systems?
  - Database Management Systems (DBMS) are software systems used to store, retrieve, and run queries on data.
  - A DBMS serves as an interface between an end-user and a database, allowing users to create, read, update, and delete data in the database.



DBMS / Database Approach



What is a Database?

- A large , shared repository of logically related data.
- A system that stores data and offers the following.
  - Ability to retrieve data
  - Ability to insert new data
  - Ability to delete unnecessary data
  - Ability to update old data

What is Stored on a Database ?

- **Data**
- **Data about data (metadata/system catalog)**
  - Metadata is a set of data that describes and gives information about other data.
  - Includes name, length, valid values, and description of a data element. Metadata is stored in a data dictionary and repository.
- Enables users to use data without worrying about how it is stored.
- Results in program data independence

What is Stored in a Database?
[Continued..]

emlployee_id	first_name	last_name	nin	department_id
44	Simon	Martinez	HH 45 09 73 D	1
45	Thomas	Goldstein	SA 75 35 42 B	2
46	Eugene	Cornelsen	NE 22 63 82	2
47	Andrew	Petculescu	XY 29 87 61 A	1
48	Ruth	Stadick	MA 12 89 36 A	15
49	Barry	Scardelis	AT 20 73 18	2
50	Sidney	Hunter	HW 12 94 21 C	6
51	Jeffrey	Evans	LX 13 26 39 B	6
52	Doris	Berndt	YA 49 88 11 A	3
53	Diane	Eaton	BE 08 74 68 A	1
54	Bonnie	Hall	WW 53 77 68 A	15
55	Taylor	Li	ZE 55 22 80 B	1

Data

Metadata

Column	Data Type	Description
emlployee_id	int	Primary key of a table
first_name	nvarchar(50)	Employee first name
last_name	nvarchar(50)	Employee last name
nin	nvarchar(15)	National Identification Number
position	nvarchar(50)	Current postion title, e.g. Secretary
department_id	int	Employee deparmtnet. Ref: Departmetns
gender	char(1)	M = Male, F = Female, Null = unknown
employment_start_date	date	Start date of employment in organization.
employment_end_date	date	Employment end date. Null if employee sti

Examples of Database Applications

- Inventory control systems
- Holiday reservation systems
- Library management systems
- Online purchasing systems
- Airline reservation systems
- Accounting systems
- Using the internet(Eg: Google, YouTube, Facebook etc.)

Advantages of Databases

Disadvantages of Databases

- Complex piece of software
- Large storage requirements and associated costs
- May be slower than retrieving from files.
- System failure and its implications

File-Based Systems Vs. Databases

File-Based	Databases
Data is scattered every where.	Data is integrated.
Sometimes different formats are used.	Data is stored in a common format for all applications to use.
The same data maybe stored more than once - duplicated.	Data duplication is reduced.
	Data can be used simultaneously by multiple users

References and Further Reading

- T. CONNOLLY & C. BEGG. Database Systems – A Practical Approach to Design, Implementation and Management, 4th Edition. Addison-Wesley, 2005.

- T. CONNOLLY & C. BEGG. Database Solutions – A step-by-step guide to building databases, 2nd Edition. Addison-Wesley, 2004.